Error: We try to estimate physical quantities either directly using instruments or by calculating based on actual measurements of some physical quantities, since most measured physical quantities can't be known to infinite decimals, there is a difference between actual value of a physical quantity and our estimation. This difference is known as error. Mathematically:

$$\text{Error} = \text{Measured Value} - \text{True Value}.$$

In most of the cases true value is not known, we can just have an estimate of true value based on averages of measured values and take that as a best estimate of true value. Mathematically:

$$\text{True Value} = \text{Measured Value} - \text{Error}$$

Also we define correction as negation of Error.

$$\therefore \quad \text{True Value} = \text{Measured Value} + \text{Correction}.$$

Resolution Of An Instrument: The smallest amount of input signal change that an instrument can detect reliably is called resolution of the instrument. For most of the instruments, it is taken as the least count of the instrument though it could be lesser due to random noise within the instrument. (For instance the last digit on a digital display could randomly vary due to internal noise within system).

[Think a computer measuring time required to copy files! ☺]

Accuracy And Precision Of A Measurement: Accuracy of a measurement refers to closeness of a measurement to the true value of the measured quantity (measurement will be mostly the average of several readings taken for estimating a physical quantity). Precision tells us the limit of resolution of the measuring instrument (For instance a typical pair of vernier calipers reads up to 0.1 mm whereas a ruler reads up to 1mm thus vernier calipers are most precise than rulers).

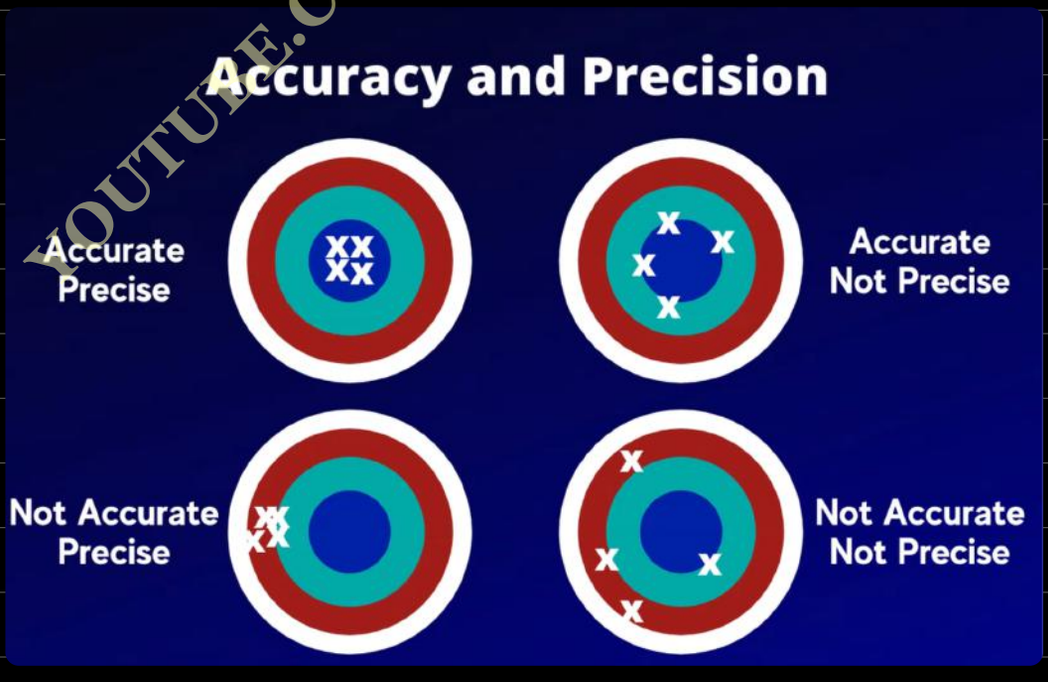
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Example: True value of a length is 3.678 cm. In one experiment using an instrument of resolution 0.1 cm, the measured value is found to be 3.5 cm while using another instrument of a greater resolution, (0.01 cm) the length is determined to be 3.38 cm. Which measurement has more accuracy? Which one has higher precision?

Sol: 3.5 cm : More Accurate [Closer to true Value]
3.38 cm : More precise [Instrument has resolution of 0.01 cm].

→ Accuracy of an instrument can depend on its resolution, for instance a vernier calipers without a zero error will be more accurate than a ruler without zero error. A ruler may give a measurement as 5.1 cm whereas a vernier may give a value closer to the true value as 5.13 cm.

→ Accuracy and Precisions of a measuring instrument can be understood from the following analogy [Think of instrument trying to find Coordinates of the Bull's eye]:



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Example: Three students are asked find length of a rod whose length is known to be 2.250 cm. Comment on precision and accuracy of their measurements:

Student	Measurement-1	Measurement-2	Measurement-3	Average
A	2.25cm	2.27cm	2.26cm	2.26cm
B	2.252cm	2.250cm	2.251cm	2.251cm
C	2.250cm	2.250cm	2.250cm	2.250cm

Solution: Comparatively:

A: Neither Accurate Nor Precise
B: More Accurate but less precise (Measurement Set)
C: Accurate and Precise. (No spread)

NATURE OF ERRORS:

(I) Systematic Errors: The errors that tend to be in one direction i.e either positive or negative are called systematic errors.
Some sources of systematic errors:

(a) Instrumental Errors: Calibration Error, Zero Error, Design Imperfection.
Examples:
Calibration Error: A thermometer could be wrongly calibrated to read 104°C at boiling point of water.
Zero Error: The reading of an instrument when it is actually supposed to give zero is called zero errors. For instance at a sweets shop, the balance might not give zero even when no sweets are placed on it (either due to wear and tear losses or due to something permanently sticking on the weighing pan).
Design Imperfection: e.g. Backlash error [In a screw gauge there might be some "play" between the screw threads

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[Nut bolt arrangement] due to which, for same amount of rotation, linear position of screw may vary slightly. [This can be minimized by rotating the screw in a single direction]. [Backlash error is both a systematic and Random error].

(b) Imperfection in measuring technique: Eg. Estimating temperature of human body by placing thermometer under armpits will always give a lower temperature. Also measurable changes in external conditions during an experiment (changes in temperature, wind velocity, humidity etc.) can cause systematic errors in the measurements.

(c) Personal Errors: Due to individual's bias (e.g. Parallax Error) on improperly setting up the instrument. (on randomly choosing upper or lower meniscus in a burette).

Parallax Error: Usually in instruments with measuring needle indicators, the plane of needle movement is above the plane of measurement etchings. Thus if we do not look along normal of these planes, the needle might appear to coincide with a different mark. Also some experiments require coinciding two points, however even without actually coinciding there can be an illusion of coinciding from a particular point of view. This is also parallax error (especially common in optics instruments).

Systematic Errors Can be Reduced By:

1. Improving Experimental Techniques
2. Using Better instruments
3. Applying appropriate corrections by estimating systematic errors (e.g. zero error can be subtracted).

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RANDOM ERRORS: Errors which are irregular with respect to sign and/or size are called random errors. They can arise due to random unpredictable fluctuations in experimental conditions (Fluctuations in Voltage, Temperature, Mechanical Vibrations etc), or personal errors.

Random errors can be generally reduced by repeating the experiments a number of times and then taking the averages. The random error reduces with number of observations ($\propto 1/N$).

LEAST COUNT ERROR: (Both Systematic and Random): No instrument can measure a continuous physical quantity to infinite number of decimals, but it can only provide its estimate within certain limits. (e.g. Rounded off to first decimal, or changing in steps of 0.5 mm). The error occuring due to resolution of an instrument is called least count error. It is considered both systematic and random because its magnitude is limited but sign is arbitrary in general. The least count error in measured value is reported as $a \pm \Delta a$ where Δa is the least count of the instrument.

GROSS ERROR (Blunder): An error occuring due to a major carelessness such as a mistake in recording or reading an instrument can be called a gross error. A gross error can't be adjusted by applying any correction factor.

ABSOLUTE ERROR, RELATIVE ERROR and % ERROR: Let $a_1, a_2, a_3 \ldots a_n$ be measurements of some physical quantity under repeated trials then the best possible value of the quantity is accepted as

$$a_{mean} = \frac{a_1 + a_2 + a_3 + \cdots a_n}{n}$$

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or $a_{mean} = \frac{\Sigma a_i}{n}$ This is because if there are random errors, we are as likely to overestimate the true value as underestimating. Taking average of a large numbers of measurements removes this bias to some degree.

ABSOLUTE ERROR: Magnitude of difference between true value of the quantity and individual measurement is called the absolute error of the measurement. We denote it by $|\Delta a|$. If there is no way of knowing the true value, a_{mean} is taken as true value and individual errors in measurements is calculated as:

$$\Delta a_1 = a_1 - a_{mean}$$
$$\Delta a_2 = a_2 - a_{mean} \quad \text{etc.}$$
$$\Delta a_n = a_n - a_{mean} \quad \text{etc.}$$

MEAN ABSOLUTE ERROR: Arithmetic mean of all absolute errors is taken as mean absolute error, it is represented by Δa_{mean}. Mathematically:

$$\Delta a_{mean} = \Sigma (|\Delta a_i|)/n.$$

Now the true value of the measurement is likely to be in the range $a = a_{mean} \pm \Delta a_{mean}$ and this is an acceptable way of reporting scientific measurements for many purposes.

[Depending on the purpose there exist other more sophisticated ways of specifying uncertainty and confidence intervals in measurements, but for JEE purpose we are happy with the method presented above ☺]

RELATIVE ERROR (or fractional error): $\text{Relative Error} = \frac{\Delta a_{mean}}{a_{mean}}$

PERCENTAGE ERROR: $\frac{\Delta a_{mean}}{a_{mean}} \times 100.$

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NCERT EXAMPLE:

Example 2.6 Two clocks are being tested against a standard clock located in a national laboratory. At 12:00:00 noon by the standard clock, the readings of the two clocks are :

	Clock 1	Clock 2
Monday	12:00:05	10:15:06
Tuesday	12:01:15	10:14:59
Wednesday	11:59:08	10:15:18
Thursday	12:01:50	10:15:07
Friday	11:59:15	10:14:53
Saturday	12:01:30	10:15:24
Sunday	12:01:19	10:15:11

If you are doing an experiment that requires precision time interval measurements, which of the two clocks will you prefer ?

Solution: For clock 1, range of variation over 7 days is 162 seconds whereas only 32 seconds for second clock. Thus first clock has more tendency of random error (even though it has smaller zero errors). However to measure time intervals we subtract initial value of time from final ∴ zero errors does not matter. ∴ for measuring time intervals second clock is preferable due to smaller tendency of random errors.

NCERT EXAMPLE:

Example 2.7 *We measure the period of oscillation of a simple pendulum. In successive measurements, the readings turn out to be 2.63 s, 2.56 s, 2.42 s, 2.71s and 2.80 s. Calculate the absolute errors, relative error or percentage error.*

Solution: $T_{mean} = (2.63 + 2.56 + 2.42 + 2.71 + 2.80)/5 = 13.12/5$

$\Rightarrow T_{mean} = 2.624\ s = 2.62\ sec$ ← ROUND OFF TO LEAST COUNT

∵ All periods are measured to a resolution of 0.01 sec ∴ T_{mean} is also reported to second decimal.

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Now we shall find individual errors in measurements:

$\Delta T_1 = 2.63\,s - 2.62\,s = 0.01\,s$

$\Delta T_2 = 2.56\,s - 2.62\,s = -0.06\,s$

$\Delta T_3 = 2.42\,s - 2.62\,s = -0.20\,s$

$\Delta T_4 = 2.71\,s - 2.62\,s = 0.09\,s$

$\Delta T_5 = 2.80\,s - 2.62\,s = 0.18\,s.$

Thus $\Delta T_{mean} = (0.01 + 0.06 + 0.2 + 0.09 + 0.18)\,sec / 5$

$\Rightarrow \Delta T_{mean} = 0.11\,sec.$ Thus the time period of pendulum must be reported as:

$$T = T_{mean} \pm \Delta T_{mean}.$$

$\Rightarrow$ $T = 2.62 \pm 0.11\,sec$ ← CORRECT METHOD.

Now the quantity reported $T \in (2.51, 2.73)$, since the mean of all absolute errors was 0.11 sec, there is already an error in tenth of a second so there is no point in reporting the result to a hundredth. A more correct way of writing it is:

$T = 2.6 \pm 0.1\,sec$ ← MORE CORRECT WAY.

Here the digit 6 is uncertain (it could be anywhere between 5 and 7) whereas we are certain about digit 2. Thus we say that the measurement has two significant figures. (The number of digits that are certain + the last digit which may have an error of ±1).

RELATIVE ERROR: $\frac{0.1}{2.6} = 0.038 \approx 0.04$

PERCENTAGE ERROR: $\frac{0.1}{2.6} \times 100 = 3.8\% \approx 4\%$.

(Sometimes percentage error is rounded off to higher value so

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you may have to mark the nearest / best matching option rather than looking for exact match. (Most of the times it is better to report % error on higher side).

ERROR OF A SUM OR DIFFERENCE : Let A and B be two physical quantities with measured values as $A \pm \Delta A$ and $B \pm \Delta B$ respectively then if $Z = A + B$ is another physical quantity then, by addition:

$$Z \pm \Delta Z = (A \pm \Delta A) + (B \pm \Delta B)$$

Maximum possible error in Z is $\Delta A + \Delta B$, thus ΔZ is taken as $\boxed{\Delta Z = \Delta A + \Delta B}$

If instead, $Z = A - B$ then:

$Z \pm \Delta Z = (A \pm \Delta A) - (B \pm \Delta B)$, once again maximum possible error in Z is $(\Delta A + \Delta B)$ (∵ error on A could be positive and error in B could be negative). For example:

$$(2.1 \pm 0.1) - (1.1 \pm 0.1) = 1.0 \pm 0.2$$

Since the value could vary from a max of $(2.2 - 1)$ to $(2 - 1.2)$ i.e (1.2 to 0.8).

Thus even in subtraction, $\pm \Delta Z = \pm (\Delta A + \Delta B)$.
(Errors are not subtracted even in a subtraction operation between measurements, they are still added).

→ When two quantities are added or subtracted, the absolute error in the final result is the sum of absolute errors in individual quantities.

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NCERT EXAMPLE:

Example 2.8 The temperatures of two bodies measured by a thermometer are $t_1 = 20\ ^0C \pm 0.5\ ^0C$ and $t_2 = 50\ ^0C \pm 0.5\ ^0C$. Calculate the temperature difference and the error therein.

Sol:

$t_2 - t_1 = (50 - 20) \pm (0.5 + 0.5)$

$= (30 \pm 1)\ ^0C$

ERROR IN A PRODUCT OR QUOTIENT:

Let $Z = AB \Rightarrow Z \pm \Delta Z = (A \pm \Delta A)(B \pm \Delta B)$

$$Z \pm \Delta Z = AB\left(1 \pm \frac{\Delta A}{A}\right)\left(1 \pm \frac{\Delta B}{B}\right) = AB\left[1 \pm \frac{\Delta A}{A} \pm \frac{\Delta B}{B} \pm \frac{\Delta A \Delta B}{AB}\right]$$

Dividing LHS by Z & RHS By AB:

$$1 \pm \frac{\Delta Z}{Z} = 1 \pm \frac{\Delta A}{A} \pm \frac{\Delta B}{B} \pm \frac{\Delta A \Delta B}{AB}$$

The last term is a kind of "second order" term

$$\Rightarrow \frac{\Delta Z}{Z} = \frac{\Delta A}{A} + \frac{\Delta B}{B}$$

Thus in product relation errors get added up.

We could also prove this by taking log and differentiating:

$$Z = AB \Rightarrow \ln Z = \ln A + \ln B$$

$$\Rightarrow \frac{dZ}{Z} = \frac{dA}{A} + \frac{dB}{B} \Rightarrow \boxed{\frac{\Delta Z}{Z} \approx \frac{\Delta A}{A} + \frac{\Delta B}{B}}$$

(Correct to the extent ΔA & ΔB are small)

Similarly if $Z = A/B \Rightarrow \ln Z = \ln A - \ln B$

$$\Rightarrow \frac{dZ}{Z} = \frac{dA}{A} - \frac{dB}{B} \Rightarrow \frac{\Delta Z}{Z} = \frac{\Delta A}{A} - \frac{\Delta B}{B}$$

However since ΔB could be either positive or negative $\therefore$

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to find maximum possible error the negative sign must be replaced by a +

$$\Rightarrow \boxed{\frac{\Delta z}{z} = \frac{\Delta A}{A} + \frac{\Delta B}{B}}$$

→ Whenever two quantities are multiplied or divided, the relative error in result is sum of relative errors in the multipliers.

→ Word of Caution: CORRELATED ERRORS:

If we have a function of the form $\frac{f(a,b)}{g(a,c)}$ we can see that both f and g depend on variable a, here we can't simply add the fractional errors in f and g. We need to see the overall effect of error in a (could have a compensating effect).

Example: (How not to add error):

$$\text{Say: } z = \frac{f}{g} = \frac{ab}{ac}, \quad \frac{\Delta f}{f} = \frac{\Delta a}{a} + \frac{\Delta b}{b}$$

$$\frac{\Delta g}{g} = \frac{\Delta a}{a} + \frac{\Delta c}{c}, \quad \frac{\Delta z}{z} = \frac{\Delta f}{f} + \frac{\Delta g}{g} = \frac{2\Delta a}{a} + \frac{\Delta b}{b} + \frac{\Delta c}{c} \leftarrow \text{Blunder}$$

Correct way: $z = \frac{f}{g} \Rightarrow \ln z = \ln f - \ln g = \ln ab - \ln ac$

$$\Rightarrow \frac{dz}{z} = \frac{da}{a} + \frac{db}{b} - \left(\frac{da}{a} + \frac{dc}{c}\right)$$

Now Combine differentials coefficients :

$$da\left[\frac{1}{a} - \frac{1}{a}\right] + db\left[\frac{1}{b}\right] + dc\left[\frac{-1}{c}\right]$$

Now Simplify before taking Mod Error.

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$\Rightarrow \frac{dz}{z} = \frac{db}{b} - \frac{dc}{c}$, Now maximize error taking mod

$\Rightarrow \frac{\Delta z}{z} = \frac{\Delta b}{b} + \frac{\Delta c}{c}$ ← Correct Result.

Another Example: $z = \frac{a}{a-b} = \frac{f}{g}$ where $f = a$, $g = a-b$.

WRONG METHOD: $\frac{\Delta z}{z} = \left(\frac{\Delta f}{f}\right) + \left(\frac{\Delta g}{g}\right) = \frac{\Delta a}{a} + \frac{\Delta(a-b)}{a-b}$

$\Rightarrow \frac{\Delta z}{z} = \frac{\Delta a}{a} + \frac{\Delta a + \Delta b}{a-b}$ (to maximize error)

$\Rightarrow \frac{\Delta z}{z} = \Delta a\left[\frac{1}{a} + \frac{1}{a-b}\right] + \frac{\Delta b}{a-b}$ ← Wrong Result.

Correct Method: (Collect Δa coefficients before taking mod)

$z = \frac{a}{a-b} \Rightarrow \ln z = \ln a - \ln(a-b)$

$\Rightarrow \frac{dz}{z} = \frac{da}{a} - \left(\frac{1}{a-b}\right)(da - db)$

Now collect da coefficients (or all correlated coefficients before taking mod)

$\Rightarrow \frac{dz}{z} = da\left[\frac{1}{a} - \frac{1}{a-b}\right] + \frac{db}{a-b}$ Now take mod

$\Rightarrow \frac{\Delta z}{z} = \left|\Delta a\left[\frac{1}{a} - \frac{1}{a-b}\right]\right| + \left|\frac{\Delta b}{a-b}\right|$ Ans.

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Example [Correlated Errors]

The relative density of a material is found by weighing the body first in air and then in water, the weight in air is found to be $(10.0 \pm 0.1)N$ and weight in water is found to be $(5.0 \pm 0.1)N$. From this maximum percentage error in relative density is found to be .....%.

Sol: Let volume of material be V and its density ρ. Then:

$W_a = \rho V g$ [Weight in air], $W_w = \rho V g - \rho_w V g$ (water)

$$\Rightarrow \frac{W_a}{W_w} = \frac{\rho}{\rho - \rho_w} = \frac{\rho/\rho_w}{\rho/\rho_w - 1} = \frac{s}{s-1} \quad [\text{Specific Gravity}]$$

$$\Rightarrow s = \left(\frac{W_a}{W_a - W_w}\right) \Rightarrow \ln s = \ln W_a - \ln(W_a - W_w)$$

$$\Rightarrow \frac{ds}{s} = \frac{dW_a}{W_a} - \left(\frac{1}{W_a - W_w}\right)(dW_a - dW_w)$$

$$\Rightarrow \frac{ds}{s} = dW_a\left[\frac{1}{W_a} - \frac{1}{W_a - W_w}\right] + dW_w\left[\frac{1}{W_a - W_w}\right]$$

$$\Rightarrow \frac{\Delta s}{s} = \left|\Delta W_a\left[\frac{1}{W_a} - \frac{1}{W_a - W_w}\right]\right| + \left|\Delta W_w\left[\frac{1}{W_a - W_w}\right]\right|$$

$$\Rightarrow \frac{\Delta s}{s} = \left|0.1\left[\frac{1}{10} - \frac{1}{5}\right]\right| + \left|0.1\left[\frac{1}{5}\right]\right| = 0.03$$

$\Rightarrow$ Relative error $= 0.03 = 3\%$ Ans.

Method II (Also correct): $s = \frac{W_a}{W_a - W_w} \Rightarrow \frac{1}{s} = 1 - \frac{W_w}{W_a}$

$$\Rightarrow -\frac{1}{s^2}ds = \frac{W_a dW_w - W_w dW_a}{W_a^2} \Rightarrow \frac{ds}{s} = -s\left[\frac{W_a dW_w - W_w dW_a}{W_a^2}\right]$$

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Now we can individually maximize the terms containing dW_w and dW_a

$$\Rightarrow \frac{\Delta S}{S} = S\left[\frac{W_a \Delta W_w + W_w \Delta W_a}{W_A^2}\right] = S\left[\frac{\Delta W_w}{W_A} + \frac{W_w}{W_A^2}\Delta W_a\right]$$

Now $S = \frac{W_a}{W_a - W_w} = 2.0 \Rightarrow \frac{\Delta S}{S} = 2.0\left[\frac{0.1}{10} + \frac{5}{100} \times 0.1\right]$

$= 0.03 = 3\%$ Ans.

WRONG METHOD: $S = \frac{W_a}{W_a - W_w} \Rightarrow \frac{\Delta S}{S} = \frac{\Delta f}{f} + \frac{\Delta g}{g}$

Where $f = W_a$, $g = W_a - W_w$. Now $\frac{\Delta f}{f} = \frac{0.1}{10}$, $\frac{\Delta g}{g} = \frac{\Delta W_a + \Delta W_w}{W_a - W_w}$

$\Rightarrow \frac{\Delta g}{g} = \frac{0.1 + 0.1}{5} = \frac{0.4}{10}$ $\quad \text{Error} = \frac{\Delta f}{f} + \frac{\Delta g}{g}$

$\Rightarrow \frac{\Delta S}{S} = 0.04 + 0.01 = 0.05 = 5\%$ ← WRONG.

EXAMPLE: In an experiment to find f using the formula: $f = uv/(u+v)$, find the percentage error in f if: $u = 80 \pm 2$ cm, $v = 40 \pm 2$ cm.

Solution: Method I: $f = \frac{uv}{u+v} = \frac{80 \times 40}{80+40} = 80/3$ cm

Now $\frac{1}{f} = \frac{1}{u} + \frac{1}{v} \Rightarrow -\frac{1}{f^2} df = -\frac{1}{u^2} du - \frac{1}{v^2} dv$

$\Rightarrow \frac{\Delta f}{f} = f\left(\frac{\Delta u}{u^2} + \frac{\Delta v}{v^2}\right) = \frac{80}{3}\left[\frac{2}{80^2} + \frac{2}{40^2}\right] = \frac{1}{24}$

$\Rightarrow$ % error: $100 \times \frac{1}{24} = \frac{25}{6}\%$ Ans.

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Method II: $\ln f = \ln u + \ln v - \ln(u+v)$

$$\Rightarrow \frac{1}{f}df = \frac{1}{u}du + \frac{1}{v}dv - \left(\frac{1}{u+v}\right)(du+dv)$$

$$\Rightarrow \frac{\Delta f}{f} = \left|\Delta u\left[\frac{1}{u} - \frac{1}{u+v}\right]\right| + \left|\Delta v\left[\frac{1}{v} - \frac{1}{u+v}\right]\right|$$

$$= 2\left[\frac{1}{80} - \frac{1}{120}\right] + 2\left[\frac{1}{40} - \frac{1}{120}\right] = \frac{1}{120} + \frac{1}{30} = \frac{1}{24}$$

$$\Rightarrow \%\ \text{error} = \frac{1}{24} \times 100 = \frac{25}{6}\ \%$$

Think Yourself: Find % error in calculation of acceleration of an atwood machine in terms of $m_1 \pm \Delta m_1$, $m_2 \pm \Delta m_2$ $(m_1 > m_2)$ and $g \pm \Delta g$.

Solution: $a = \frac{(m_1 - m_2)}{(m_1 + m_2)}g \Rightarrow \frac{da}{a} = \left(\frac{dm_1 - dm_2}{m_1 - m_2}\right) - \left(\frac{dm_1 + dm_2}{m_1 + m_2}\right) + \frac{dg}{g}$

$$\frac{\Delta a}{a} = \left|\Delta m_1\left[\frac{1}{m_1 - m_2} - \frac{1}{m_1 + m_2}\right]\right| + \left|\Delta m_2\left[\frac{1}{m_1 - m_2} + \frac{1}{m_1 + m_2}\right]\right| + \left|\frac{\Delta g}{g}\right|$$

Ans.

NCERT EXAMPLE: Two resistances are given as $R_1 = 100 \pm 3\,\Omega$ and $200 \pm 4\,\Omega$. Find the resistance of their parallel combination with appropriate % error and absolute error.

Sol: $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} \Rightarrow -\frac{1}{R^2}dR = -\frac{1}{R_1^2}dR_1 - \frac{1}{R_2^2}dR_2$

$$\Rightarrow \frac{dR}{R} = R\left[\frac{1}{R_1^2}\Delta R_1 + \frac{1}{R_2^2}\Delta R_2\right], \quad R = \frac{R_1 R_2}{R_1 + R_2} = \frac{200}{3}\,\Omega$$

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$$\Rightarrow \frac{\Delta R}{R} = \frac{200}{3}\left[\frac{3}{100^2} + \frac{4}{200^2}\right] = \frac{2}{75} \Rightarrow \Delta R = \frac{200}{3} \times \frac{2}{75} = \frac{16}{9}\,\Omega$$

$$\Rightarrow \%\ \text{error} = \frac{2}{75} \times 100 = 2.67\,\%$$

Now $\frac{16}{9} = 1.\overline{7}$ and $\frac{200}{3} = 66.6\overline{7}$, Thus resistance can lie between, $66.6\overline{7} \pm 1.\overline{7}$ i.e. $68.\overline{4}$ and $64.\overline{8}$

Thus we can write resistance as $66.7 \pm 1.8\,\Omega$

ERROR IN A MEASURED QUANTITY RAISED TO POWER:

Suppose $Z = A^2 \Rightarrow \ln Z = 2\ln A$

$\Rightarrow \frac{\Delta Z}{Z} = 2\frac{\Delta A}{A}$. Similarly in general:

If $Z = A^p B^{q/C^r}$ then $\frac{\Delta Z}{Z} = p\frac{\Delta A}{A} + q\frac{\Delta B}{B} + r\frac{\Delta C}{C}$

(If A, B and C are not correlated Quantities).

→ Rule: The relative error in a physical quantity raised to power k is k time relative error in the quantity.

NCERT EXAMPLE:

▶ **Example 2.12** The period of oscillation of a simple pendulum is $T = 2\pi\sqrt{L/g}$. Measured value of L is 20.0 cm known to 1 mm accuracy and time for 100 oscillations of the pendulum is found to be 90 s using a wrist watch of 1 s resolution. What is the accuracy in the determination of g?

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Sol: $T = 2\pi\sqrt{\frac{l}{g}} \Rightarrow g = \frac{4\pi^2 l}{T^2} \Rightarrow \frac{\Delta g}{g} = \frac{\Delta l}{l} + 2\frac{\Delta T}{T}$

$\Rightarrow \frac{\Delta g}{g} = \frac{1}{200} + 2 \times \frac{1}{90}$ ($\because$ error in 90 oscillation = ± 1 sec)

$\Rightarrow \frac{\Delta g}{g} = \frac{49}{1800} = 0.027\overline{2} \approx 0.027$, % Error = 2.7%

or % Error $\approx$ 3% [As per NCERT]

Small Errors in Arbitrary functions:

Let some quantity be $Z = f(a, b, c)$ then if errors a, b and c are small, variation in Z can be found using Taylor approximation i.e:

$$dZ = \frac{\partial f}{\partial a} da + \frac{\partial f}{\partial b} db + \frac{\partial f}{\partial c} dc$$

$$\Rightarrow \Delta Z = \left|\frac{\partial f}{\partial a} \Delta a\right| + \left|\frac{\partial f}{\partial b} \Delta b\right| + \left|\frac{\partial f}{\partial c} \Delta c\right|$$

Example: Suppose an angle is measured to be $\theta = 60° \pm 1°$ Find the error in $\cos\theta$.

Solution: $f(\theta) = \cos\theta \Rightarrow \Delta f \approx \left|\frac{\partial f}{\partial \theta} \times \Delta\theta\right| = |-\sin\theta \times \Delta\theta|$

$\Rightarrow \Delta f = \left|-\sin 60° \times 1° \times \frac{\pi}{180°}\right| = \frac{\sqrt{3}}{2} \times \frac{\pi}{180} \approx 0.015$

$\Rightarrow \cos\theta = 0.5 \pm 0.015$

CAUTION: This method may fail miserably for ill behaved functions or points where a function is ill behaved (Such as near $\theta = 90°$, very small fluctuations in angle can lead to very large errors in $\tan\theta$)

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Example: Base of a right triangle is measured to be 100 ± 1 cm and the angle θ is measured to be $60° \pm 1°$. Estimate the area of Δ with appropriate error limits.



Solution: $A = \frac{1}{2} a \times a \tan\theta = \frac{a^2}{2} \tan\theta$

$$\Rightarrow \Delta A = \frac{\partial A}{\partial a} \Delta a + \frac{\partial A}{\partial \theta} \cdot \Delta\theta = \left| a \tan\theta \, \Delta a \right| + \left| \frac{a^2}{2} \sec^2\theta \, \Delta\theta \right|$$

$$= 100 \times \sqrt{3} \times 1 + \frac{100^2}{2} \times 4 \times \frac{1}{180} \times \pi$$

$$\Rightarrow \Delta A = 522.27 \text{ cm}^2$$

$$A = \frac{1}{2} \times 100^2 \times \sqrt{3} = 8660.25 \text{ cm}^2 \Rightarrow A \pm \Delta A = 8660.25 \pm 522.27 \text{ cm}^2$$

Significant Figures: No continuous physical quantity can be determined to infinite decimal places, thus there will always be uncertain digits in any physical measurement. The reliable digits + first uncertain digit in any measurement are known as significant digits or significant figures.

Example: Suppose time period of pendulum is reported as 1.62 s we can be sure of digits 1 and 6 but 2 is uncertain ∴ no of sig figs in the number = 3.

→ If length of an object is reported as 287.5 cm then 2, 8, 7 are certain but 5 is uncertain thus no. of sig figs = 4.

→ Significant figures indicate precision of the measurement which depends on measuring instrument. For instance if a measurement is reported as 1.62 sec, it could be inferred that least count of the stopwatch is possibly 0.01 sec.

Rules For Counting Significant Figures: A choice of units or conversion to other units should not change the number of sig figs. e.g. 2.308 cm has four sig figs. However upon

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changing the units it can also be written as: 0.02308 m or 23.08 mm or 23080 μm. All these have same sigfigs 2, 3, 0 and 8, this shows that location of decimal has no importance in determining sig figs. This example gives following rules:

→ All non zero digits are significant

→ All zeroes between non zero digits are significant inspective of decimal point.

→ If the number is less than 1, all leading zeroes (to the left of first non-zero digit) are not significant e.g in 0.00 2308 the underlined zeroes are not significant.

→ Trailing zeroes in a number without a decimal point are not significant eg. 2308 00 has only four sig figs (underlined zeroes are not significant). Similarly 1230 mm has only three significant figures.

→ Trailing zeroes in a number with a decimal point are significant. e.g:
0.06900 has four sig figs: 69 and last two zeroes.
(leading zeroes are not significant).

<u>The Unavoidable Problem With Standard Decimal Notation</u>: We devised so many rules with significant figures so that their count does not change upon changing the units. However our rules discussed so far would fail us in some situations.
e.g. Let a length be 4.700 m this has four significant figures according to our rules. This can be written as 470.0 cm, still 4 sigfigs. But how to write it in mm retaining the number of significant figures.
If we write 4700 mm, there are only two significant figures
If we write 4700.0 mm it makes five significant figures.

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SCIENTIFIC NOTATION: To avoid ambiguity in counting the significant figures as seen in the previous section, a scientific notation is used where every quantity is represented as $a \times 10^b$ where a is a number between 1 and 10 and b is any positive or negative exponent of 10. Here the significant figures are counted in the number a only. This creates the possibility of retaining significant figures despite conversion of units. For instance:

$4.700 \times 10^0\ m = 4.700 \times 10^2\ cm = 4.700 \times 10^3\ mm = 4.700 \times 10^{-3}\ km$

in all of these expressions, number of significant figures is four.

ORDER AND ORDER OF MAGNITUDE: If we have a number of the form $a \times 10^b$ and $0 < a \leq 5$ then we can get a number by rounding off a to 1 similarly if $5 < a \leq 10$ the we can get a number by rounding off a to 10. This number obtained on appropriately rounding off a is called order of magnitude.

For instance: Diameter of Earth: $D = \underbrace{1.28}_{a} \times 10^{\overbrace{7}^{b}}\ m$

After rounding off a we get $1 \times 10^7\ m$, thus we say that the diameter of earth is of order $10^7\ m$ or order of magnitude 7.

Similarly, Mass of electron is $9.1 \times 10^{-31}\ kg \approx 10 \times 10^{-31}\ kg = 10^{-30}\ kg$. Thus we say mass of electron is of the order of $10^{-30}\ kg$ or order of magnitude -30.

Order of magnitude is often used to describe rough comparison of relative magnitudes of physical quantities.

For instance diameter of hydrogen atom is $1.06 \times 10^{-10}\ m$ of order of magnitude -10, thus diameter of earth is 17 orders of

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magnitude greater than diameter of hydrogen atom.

ENGINEERING NOTATION: This is somewhat similar to scientific notation, i.e number is written as $a \times 10^b$ however b is in multiples of three (or expressed as prefix for powers) e.g:

3×10^6 m or 3 Mega Metre.

Here again significant figures are counted in a using standard rules only.

EXACT MATHEMATICAL CONSTANTS: Suppose we write circumference of a circle as $l = 2\pi R$, Here 2 is not a measured physical quantity, but an exact mathematical constant known up to infinite decimals, thus we say that this '2' has infinite significant figures. Similarly π and e or $\sqrt{2}$ all can be evaluated to any number of significant figures as required. All exact constants have infinite significant figures.

MATHEMATICAL OPERATIONS WITH SIGNIFICANT FIGURES: Suppose a Bhaiyaarwala or Kabaadiwala weigh some newspapers and says they weigh 2.5 Kg and you have a diamond crystal weighing 0.012 Kg, and now we want to add the two weights, Then would you feel comfortable saying that combined weight of newspaper and diamond crystal is 2.512 Kg. Infact if we just mechanically add and report the quantity we give an undue impression that the combined weight has an accuracy of ± 0.001 Kg, which is obviously not true. Because when a Kabadiwala says a weight is 2.5 Kg, it could possibly be 2.5 ± 0.1 Kg. Thus there is considerable error in the first quantity and it must be duly taken care of while reporting the final result for scientific purpose calculations.

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ADDITION AND SUBTRACTION: In addition or subtraction of the final result should be rounded off to as many decimal places as are there in the number with least number of decimal places.

Example:

$$\begin{array}{rr} & 436.32\ g \\ + & 227.2\ g \\ + & 0.301\ g \\ \hline & 663.821\ g \end{array}$$

← Least decimal places (1 place)

← Mechanical Addition

→ Now round off to 1 decimal place ⇒ $W = 663.8\ g$ Ans

SIGNIFICANT FIGURES IN MULTIPLICATION OR DIVISION: Suppose mass of an object were $4.237\ g$ and its volume were $2.51\ cm^3$ then by dividing using a calculator, ratio of mass and density is output as $1.688047808764\ g/cm^3$, up to 11 decimals. However reporting density to such high precision would be absurd because the quantities from which it is calculated themselves have much lesser precision. Thus we have the rule:

In Multiplication or Division final result should have as many significant figures as in number with least significant figures.

e.g.

$$\rho = \frac{m}{V} = \frac{4.237\ g}{2.51\ cm^3} = 1.68804\ g/cm^3$$

($4.237\ g$ ← 4 Sigfigs; $2.51\ cm^3$ ← 3 Sigfigs; $1.68804\ g/cm^3$ ← Mechanical Calculation)

Now round off to three sigfigs: $1.68804 \rightarrow 1.69\ g/cm^3$ Ans

INTERMEDIATE CALCULATION RESULTS: To avoid accumulation of round off errors in multistep calculations, the intermediate steps should be calculated to one extra significant figure than the standard rules. e.g. $\frac{1}{9.58} = 0.104$ to 3 significant figures,

but $\frac{1}{0.104} = 9.62$, However if we write $\frac{1}{9.58} = 0.1044$

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and then two $\frac{1}{0.1044} \simeq 9.578 = 9.58$ to three sig. figs.

NCERT EXAMPLE: Edge of cube is 7.203 m. Find its surface area and volume to appropriate sig. figs.

Sol: $S = 6l^2 = 6 \times (7.203^2) \simeq 311.299 = 311.3\ m^2$
(Note 6 has infinite sig. figs.).

$V = l^3 = 7.203^3 \simeq 373.714 \simeq 373.7\ m^3$ to correct sig. figs.

EXAMPLE: $m = 5.74g$, $V = 1.2\ cm^3$, $\rho = ?$

Sol $\rho = m/V = 5.74/1.2 = 4.7833 \simeq 4.8\ g/cm^3$ to correct sig figs.

ERROR IN A QUANTITY EXPRESSED IN SIGNIFICANT FIGURES: Unless explicity stated, the error in a quantity expressed in significant figure is taken as ± 1 in the last significant place.

E.g. $16.2\ cm \Rightarrow 16.2 \pm 0.1\ cm$

$2.99\ ms^{-1} \Rightarrow 2.99 \pm 0.01\ ms^{-1}$

Once the error is written in this form, the errors in operations can be calculated as per error propagation theory discussed earlier.

EXAMPLE: $l = 16.2\ cm$, $b = 10.1\ cm$ for a rectangle, report the area with appropriate error limits.

Sol: $A = l \times b \Rightarrow \frac{\Delta A}{A} = \frac{\Delta l}{l} + \frac{\Delta b}{b} = \frac{1}{162} + \frac{1}{101} = 1.61\%$

$A = 16.2 \times 10.1 = 163.62 \Rightarrow A \pm \Delta A = 163.62 \pm 1.61\%$

$\Rightarrow A \pm \Delta A = 163.62 \pm 2.63\ cm^2$

$\Rightarrow A \pm \Delta A = 164 \pm 3\ cm^2$ ($\because$ units digit itself is uncertain).

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→ Note that % error depends not only on number of sig. figs but also the number itself.

eg. error in $16.2\ cm = \frac{1}{162} \times 100$ while in $10.1\ cm \rightarrow \frac{1}{101} \times 100$.

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